

BASIC LEVEL III PRACTICE EXAM

1. According to SNT-TC-1A, certification is deemed revoked when the certified individual:
 - a. Is off sick for more than one week
 - b. Makes a mistake in a test report
 - c. Leaves the employer
 - d. uses different instrumentation than used during certification

13.1

2. According to SNT-TC-1A, an employer's written practice for administration of NDT personnel training, examination and certification:

- a. Should reflect the guidelines set forth in SNT-TC-1A paragraph 1, SCOPE
- b. Should be prepared by quality control management
- c. Should be reproduced verbatim from SNT-TC-1A
- d. Should be submitted to ASNT for approval
- e. Is a mandatory requirement of the US government

1.0

3. According to SNT-TC-1A, recertification may be based upon:

- a. No documented errors in interpretation
- b. Evidence of continued satisfactory performance or re-examination
- c. Quantity of noncompliance reported
- d. Contributions to ASNT

12.0

4. According to SNT-TC-1A, the recommended composite passing grade for examinations is:

- a. 70%
- b. 80%
- c. 90%
- d. 95%

8.1.5

BASIC LEVEL III PRACTICE EXAM

5. According to SNT-TC-1A, the recommended passing grade considered for each individual written examination is:

- a. 70%
- b. 80%
- c. 90%
- d. 95%

8.1.5

6. According to SNT-TC-1A, NDT Level I and II practical examinations should be administered and graded by:

- a. NDT Level III's only
- b. Any company official not directly responsible for NDT activities
- c. An NDT Level III individual..3
- d. Persons designated as an NDT examiner

8.1.2.1

7. According to SNT-TC-1A, an individual may fail a practical exam by not getting a sufficient score, or

- a. Failing to achieve the required composite score
- b. There is no other means of failing an examination
- c. By challenging questions
- d. Failure to complete required check points

8.5.4.3 and 8.5.5.3

8. The purpose of SNT-TC-1A is to:

- a. Establish guidelines for the qualification and certification of NDT personnel
- b. Establish mandatory rules for qualification and certification of NDT personnel
- c. Be incorporated into QA programs for qualification of NDT personnel
- d. B and C are true for ASME programs while answer A is correct for all others.

1.1

9. According to SNT-TC-1A, certification is defined as:

- a. Verification of qualification
- b. Notarized validation of qualification
- c. Written testimony of qualification
- d. Level I, II, III and Level II limited training records

2.1.2.

10. Which of the following is not specifically addressed by SNT-TC-1A as a method?

- a. Acoustic Emission
- b. Thermal Infrared
- c. Visual
- d. Bubble
- e. Leak

Table 6.3.1 A

11. According to SNT-TC-1A, while in the process of being qualified and certified to NDT Level I, an individual should be considered a:

- a. Trainee
- b. Instructor
- c. Candidate
- d. Level 1 Limited
- e. Assistant NDT Technician

4.2

12. According to SNT-TC-1A, the employer shall establish a written practice for:

- a. Control and administration of NDT personnel training
- b. Control of NDT personnel examinations
- c. Control of NDT personnel certification
- d. All of the above

5.0

BASIC LEVEL III PRACTICE EXAM

13. According to SNT-TC-1A, documented training and/or experience gained in positions and activities equivalent to those of Level I, II, or III prior to establishment of the employers written practice may be:

- a. Considered for certification
- b. Not be considered for certification
- c. Considered only if ASNT has approved
- d. None of the above apply

6.2

14. According to SNT-TC-1A, NDT examinations should be administered and graded by:

- a. NDT Level III's only
- b. Any company official not directly responsible for NDT activities
- c. An NDT Level III individual or designated representative
- d. Persons designated as an NDT examiner

8.1.2

15. According to SNT-TC-1A, vision examinations should assure natural or corrected near distance acuity in at least one eye such that the individual is capable of reading a minimum of Jaeger Number _____ at 12 inches.

- a. 1
- b. 2
- c. 3
- d. 4
- e. 8

8.2.1

16. According to SNT-TC-1A, the vision examination should demonstrate the capability to:

- a. Distinguish red and green for contrast analysis
- b. Differentiate contrast
- c. Perceive rated contrast differences as measured by the Ishihara plates
- d. Distinguish between the colors or shades of gray used in the method as determined by the employer

8.2.2

17. When an employer engages an outside agency to provide NDT Level III services, the responsibility for certification is:

- a. Retained by the vendor providing the services
- b. Retained by a third party
- c. Retained by the employer
- d. Can be any of the above as stated in the employer's written practice

7.4, 8.1.6, and 9.3

18. Personnel failing to attain a passing grade on any one exam:

- a. May be certified after re-evaluation by the NDT Level III
- b. Must wait 90 days and show evidence of retraining prior to reexamination and certification
- c. May be certified based upon experience and training rather than examination
- d. Must wait at least 30 days or show evidence of having received additional training as determined by the employer before reexamination

8.7 SNT-TC-1A

19. According to SNT-TC-1A, the composite grade for NDT qualification examinations must be:

- a. a simple average of 80% minimum
- b. developed by applying a weighting to each examination
- c. based upon a historical curve
- d. as prescribed in the employer's written practice

1.4, 8.1.4 and 8.1.5

BASIC LEVEL III PRACTICE EXAM

20. According to SNT-TC-1A, personnel certifications and copies of the written practice shall be maintained on file by:

- a. The certified individual
- b. The certifying agency
- c. The NDT Level III examiner
- d. The employer

9.4

21. According to SNT-TC-1A, all levels of NDT personnel shall be recertified:

- a. Every year
- b. Every 3 years
- c. Every 5 years
- d. As specified in the employer's written practice

12.2 and 1.4

22. According to SNT-TC-1A, NDT personnel may, at any time:

- a. Be reexamined
- b. Have their certifications extended
- c. Have their certifications revoked
- d. All of the above

10.1

23. According to SNT-TC-1A, when employment is terminated:

- a. Certifications shall be deemed revoked
- b. Certifications shall be considered valid for that period specified in the employer's written practice
- c. Certifications shall be considered suspended
- d. Qualifications shall be deemed revoked

13.1

24. According to SNT-TC-1A, a Level I, II, or III whose certification has been terminated may be recertified to his/her former NDT Level by a new employer based upon examinations provided:

- a. The employee has proof of prior certification
- b. The employee was working in the capacity to which certified within six months of termination
- c. The employee is recertified within six months of termination
- d. All of the above must be true.

13.2

25. According to SNT-TC-1A, which of the following statements is NOT true?

- a. Certification is written testimony of qualification
- b. Qualification is the skill, training, and experience required for personnel to properly perform the duties of a specific job
- c. The employer is the corporate or private entity which employs personnel for wages or salary
- d. A certifying agency is any organization used by an employer in training NDT personnel
- e. Training is a program to impart the knowledge and skills necessary for qualification

2.1.2 -2.1.4

26. Use of SNT-TC-1A is mandatory when:

- a. The material being tested is for an USAF contract
- b. The material being tested if for an USN contract
- c. It has been specified by the customer
- d. The material being tested is for an army contract which requires certification of NDT personnel
- e. Any of the above

Forward

BASIC LEVEL III PRACTICE EXAM

27. Which of the following statements is NOT true concerning use of SNT-TC-1A?

- a. It is intended for use by any industrial segment
- b. It was not made to satisfy any specific code or standard
- c. It was last revised in 2020
- d. It is not intended to be used as a strict specification

Cover; Forward

28. According to SNT-TC-1A, should personnel that operate digital electrical conductivity or digital ultrasonic thickness gages be qualified and certified?

- a. Yes, because SNT-TC-1A requires that all personnel performing NDT be qualified and certified
- b. No, because SNT-TC-1A does not cover that specific operation
- c. Only if required by industry codes, standards, and specifications
- d. If NDT personnel should be qualified and certified depends solely upon the needs of the employer and the requirements of the employer's customers or clientele

1.4

29. Which of the following is not within the scope of activities of a Level II individual, as recommended by SNT-TC-1A?

- a. Standardize equipment
- b. Prepare written instructions
- c. Establish techniques
- d. Provide training

4.3.2 and 4.3.3

BASIC LEVEL III PRACTICE EXAM

30. According to SNT-TC-1A, may a Level I independently perform, evaluate, and record results of NDT tests with occasional supervision of a Level II or III?
- Yes, the intent in SNT-TC-1A is that the Level I may perform the above functions provided they are in accordance with written procedures and so documented in the employer's written practice
 - No, the Level I may perform the above functions in accordance with written procedures, but must be under constant supervision and guidance of a Level II or III
 - No, the Level I may not sign for results of test results
 - No, the Level I is not allowed to take any independent action
 - Yes, provided that the Level I receives detailed verbal instructions from the Level II or III and demonstrates this capability to follow the instructions

4.3.1. Inquiry 2009-2

31. According to SNT-TC-1A, when an employer uses an outside agency to provide NDT Level III services, the responsibility for certification must be:

- Assumed by the contracted agency
- Witnessed by a third party
- Retained by the employer
- Can be any of the above if specifically stated in the employer's written practice

2.1.16, 8.1.6, and 9.3

32. According to SNT-TC-1A, personnel failing to attain the required examination grades:

- Can be certified after re-evaluation by an NDT Level III
- Must wait 60 days and show evidence of re-training prior to re-examination and certification
- Can be certified based upon experience rather than written examination
- Must wait at least 30 days or show evidence of having received suitable additional training as determined by the NDT Level III before re-examination

8.7

BASIC LEVEL III PRACTICE EXAM

33. According to SNT-TC-1A, the composite grade for NDT qualification examinations may be:

- a. A simple average of the examinations
- b. Developed by applying a weighting factor to each examination
- c. Based upon a historical curve
- d. Either a or b as described in the employer's written practice

8.1.5 and 1.4

34. According to SNT-TC-1A, personnel certifications and copies of the employer's written practice shall be maintained on file:

- a. By each certified individual
- b. By the certifying agency
- c. By the NDT Level III examiner
- d. By the employer

5.6 and 9.4

35. According to SNT-TC-1A, which of the following is not true?

- a. Certification is written testimony of qualification
- b. Qualification is the skill, training, and experience required for personnel to properly perform the duties of a specific job
- c. The employer is the corporate or private entity which employs personnel for wages or salary
- d. A certifying agency is any organization used by an employer in training NDT personnel
- e. Training is a program developed to impart knowledge and skills necessary for qualification.

2.1.2, 2.1.3, 2.1.4, 2.1.9, 2.1.16, and 2.1.24

36. According to SNT-TC-1A, a candidate for Level III with a four year degree in fine arts would require how many months of experience as a Level II or equivalent to qualify for Level III?

- a. 4
- b. 12
- c. 24
- d. 36
- e. 48

6.3.2

37. According to SNT-TC-1A, records of vision examinations should be kept for:

- a. 1 year
- b. 3 years
- c. 5 years
- d. The current certification period
- e. As long as the employee is employed by the employer

9.4.5

38. According to SNT-TC-1A, what is the lowest Level that can organize and report the results of NDT tests?

- a. Trainee
- b. Level I
- c. Level II
- d. Level III

4.3.2

39. According to SNT-TC-1A, what is the lowest Level that can provide guidance to trainees?

- a. Level I
- b. Level II
- c. Level III
- d. QA Manager

4.3.2

BASIC LEVEL III PRACTICE EXAM

40. According to a contract, the employer is to meet the minimum recommendations of SNT-TC-1A. What should the employer do?

- a. Use SNT-TC-1A verbatim
- b. Use SNT-TC-1A verbatim except change each occurrence of the word "should" to "shall"
- c. Use current written practice
- d. Use SNT-TC-1A to draft a written practice and change each occurrence of the word "should" to "shall"

Forward

41. According to SNT-TC-1A, which of the following personnel may not be designated as a certifying authority for the employer?

- a. Outside agency Level III
- b. Company president
- c. Company Level III
- d. Company Janitor

2.1.3, 2.1.4, and 9.3

42. According to SNT-TC-1A, which of the following is NOT required in the certification records?

- a. Signature of Level III that verified qualifications of candidate for certification
- b. Current copies of examinations or suitable evidence of successful completion of examinations
- c. Grades obtained during training
- d. Level of certification and NDT method
- e. Results of vision examinations obtained during the current certification period

9.4

43. According to SNT-TC-1A, training documentation must contain:

- a. Statement indicating satisfactory completion of training in accordance with the employer's written practice
- b. Instructor signature of approval
- c. Statement by outside agency indicating satisfactory completion of training in accordance with the agency's written practice
- d. ASNT approval

9.4.4

44. An employer decides to reduce training hour requirements below the minimums recommended in SNT-TC-1A. Is this permissible?

- a. Yes
- b. No
- c. None of the above as SNT-TC-1A does not address training

1.4, Table 6.3.1A Note 3

45. According to SNT-TC-1A, which of the following is true?

- a. Overtime hours cannot be considered in satisfying minimum experience
- b. Overtime can be considered only if the technician is being qualified in more than one method simultaneously
- c. Overtime can be credited based upon total hours
- d. SNT-TC-1A does not address overtime

Study Guide

46. High school education or equivalent is recommended in SNT-TC-1A as a minimum requirement for:

- a. Level III only
- b. Level II and III only
- c. None of the Levels
- d. All the Levels

Study Guide and Table 6.3.1A

BASIC LEVEL III PRACTICE EXAM

47. Recommended training reference material:

- a. Is available only through ASNT
- b. Is available from a variety of sources
- c. Can only include those references listed
- d. Must be made available to each trainee
- e. Is completely contained in the NDT Handbooks published by ASNT

7.3

48. According to SNT-TC-1A, recommended NDT course outlines:

- a. Are included in CP-105 for 16 NDT methods
- b. Are included in SNT-TC-1A
- c. Are not available for vibrations and laser methods
- d. Are included in CP-105, but not for vibrations or laser methods

7.3

49. According to SNT-TC-1A, which of the following statements is true with regard to the administration of written examinations?

- a. The examinee should not be permitted access to any reference material
- b. Reference data may be supplied with practical exams only
- c. Level III personnel should be required to memorize everything except codes, specifications and procedures
- d. Codes, specifications, and procedures can be provided to examinees for reference during examinations provided that they do not contain data that can be used to answer questions in the general examination
- e. None of the above are true

8.1.3

50. According to SNT-TC-1A, the vision acuity test should be administered:

- a. Once
- b. Annually
- c. Once every three years
- d. Once every five years

8.2.1

51. According to SNT-TC-1A, the color vision test should be administered:

- a. Once
- b. Annually
- c. Once every 3 years
- d. Once every five years

8.2.2

52. According to SNT-TC-1A, color vision is adequate:

- a. Only if the candidate can read 9 out of 14 Ishihara plates accurately
- b. Only if the candidate can read 9 out of 14 pseudoisochromatic plates accurately
- c. If the candidate can distinguish between the colors used in the method
- d. If the candidate can read a J2 chart at 15 inches

8.2.2

53. Which of the following is true regarding Level III examination requirements according to SNT-TC-1A?

- a. The specific exam is required under all circumstances.
- b. The specific exam may be waived if the Level III candidate has an ASNT NDT Level III certificate in the method
- c. The procedure preparation exam may not be waived under any circumstances
- d. The procedure preparation exam is required only if the candidate does not have an ASNT NDT Level III certificate.

8.6.3.1 and 8.6.4.1

BASIC LEVEL III PRACTICE EXAM

54. According to CP-189, which of the following is true regarding recertification of Level I and II inspectors?

- a. Recertification can only be accomplished by written and practical examination
- b. Recertification is permissible by specific examination and documentation of 350 hours of experience in the method during the prior certification period
- c. Recertification can only be accomplished by maintaining a valid ACCP certificate in the method and a specific examination
- d. Re-examination by ASNT

8.1

55. According to CP-189, the vision acuity test shall be administered:

- a. Once
- b. Annually
- c. Once every three years for Level I and II, and every five years for Level III
- d. Once every five years

6.1.3 and 7.1.2

56. According to CP-189, the color vision test shall be administered:

- a. Once
- b. Annually
- c. Once every three years for Level I and II, and every five years for Level III
- d. Once every five years

6.1.3, 7.1.2 and 8.2

57. According to CP-189, color vision is adequate:

- a. Only if the candidate can read 9 out of 14 Ishihara plates accurately
- b. Only if the candidate can read 9 out of 14 pseudo-isochromatic plates accurately
- c. If the candidate can distinguish between the colors used in the method
- d. If the candidate can read a J1 chart at a distance not less than 12 inches in at least one eye

6.1.2

58. According to CP-189, which of the following is true regarding vision acuity exams:

- a. Acuity is given annually regardless of certification level, and J1 at a distance not less than 12 inches in at least one eye is required
- b. Acuity is measured at five year intervals and is measured using J2 letters at a distance not less than 12 inches
- c. Acuity is given annually regardless of certification level using J2 at a distance not less than 12 inches in at least one eye
- d. Acuity is measured at the recertification interval and the candidate must distinguish between colors used in the method.

6.1.1 and 6.1.3

59. According to the ASNT Code of Ethics, an ASNT NDT Level III who exaggerates or overstates the number of methods for which qualified would most likely be subject to the following:

- a. Appearance before the Ethics Committee and a year long suspension of the ASNT NDT Level III certificate
- b. Appearance before the Ethics Committee and reduction of certification to Level II
- c. Appearance before the ASNT Awards Committee and a seat on the National Certification Board
- d. Appearance before Ethics Committee of the National Certification Board and subject to sanctions such as a permanently revoked certificate

Code of Ethics 9.1 and 9.2 (page 119 5th ed Study Guide)

60. According to ANSI/ASNT CP-189, in what case could a Level II administer and grade a written exam of other Level I and II personnel?

- a. If approved by a Level III and is an essay type test
- b. If a multiple choice examination and delegation is approved and documented by the Level III
- c. Level II personnel cannot grade or administer exams under any circumstances due to potential conflicts of interest
- d. If the Level II has been approved by ASNT and hold a valid Level II certificate in the method.

6.4.1.3

BASIC LEVEL III PRACTICE EXAM

61. According to ANSI/ASNT CP-189, what is cause for revocation of a certification?

- a. CP-189 does not address time periods of inactivity and leaves it to the employer to set maximum limits of inactivity
- b. Not performing in the method for 12 consecutive months, the vision exam exceeds one year
- c. Not performing in the method for 24 consecutive months; for a revoked ASNT NDT Level III certificate; or unethical/incompetent conduct
- d. When the individual's performance is determined to be deficient in the method; or any time at the discretion of the employer

7.3

62. According to ANSI/ASNT CP-189, which of the following credentials are required to administer Level III exams to candidates?

- a. The administrator shall possess an NDT Level III certificate and be familiar with the specifications, standards and products of the employer
- b. The administrator shall be approved by an NDT Level III and documented
- c. The administrator has a minimum of 1 years experience in the method equivalent to a Level II and has graduated from a minimum four year college or university with a degree in engineering or science
- d. The administrator has four years experience beyond the Level II requirements in NDT in an assignment at least comparable to that of a Level II in the applicable NDT method

6.7

63. Where would you find the experience requirements necessary to take an ASNT Level III method exam?

- a. SNT-TC-1A
- b. The examination application package
- c. ANSI/ASNT CP-105 – *Topical Outlines for Qualification of NDT Personnel*
- d. Actual performance of an NDT method conducted in the work environment resulting in the acquisition of knowledge and skill. This does not include formal classroom training, but may include laboratory and on the job training as defined in the employer's certification procedure

10.1.2 and 4.2.3

BASIC LEVEL III PRACTICE EXAM

64. In accordance with ANSI/ASNT CP-189 what must be done to recertify a Level III?

- a. Every 5 years currency of employee's ASNT NDT Level III certificate shall be verified.
- b. Retest by ASNT of the method exam, followed by successful completion of specific, procedure preparation and demonstration exams administered by the employer
- c. Retest by ASNT of the method exam, followed by successful completion of specific, procedure preparation and demonstration exams administered by an outside agency
- d. Evidence of continued satisfactory performance and re-examination of those portions of the exams deemed necessary by the employers NDT Level III

8.2

65. In accordance with ANSI/ASNT CP-189, NDT procedures and personnel qualification and certification procedures must be approved by:

- a. Level III
- b. Quality Assurance Manager with ISO 9000 qualification
- c. Local union officials and management
- d. A staff member with academic credentials at least equivalent to a B.S. in engineering, physical science or technology, and possess adequate knowledge in the NDT method

2.1.17, 3.2 and 5.3

66. In accordance with ANSI/ASNT CP-189, how does an instructor become qualified?

- a. There are no specific requirements for an instructor.
- b. An instructor may have an Level III certificate; have academic credentials equivalent to a BS in engineering with science or technology with adequate knowledge in the NDT method; be a graduate of 2 year school of science and have 5 or more years experience as Level 2 or equivalent; or have 10 or more years experience as a Level 2 or equivalent
- c. Designation by Level III is required for all instructors
- d. b and c

2.1.16, 4.4.1 and 4.4.2

BASIC LEVEL III PRACTICE EXAM

67. In accordance with ANSI/ASNT CP-189, training requirements for Level III are satisfied if the candidate has:

- a. a current ASNT NDT Level III certificate in the specific NDT method
- b. 80 hours of training and 1600 hours of experience
- c. Has completed a Basic exam with a minimum of 55 questions and a Method exam with 65 questions
- d. Has completed an employer based Basic, Method, Specific and Procedure Preparation examinations

4.1.4

68. In accordance with ANSI/ASNT CP-189, what are the minimum requirements for administration of the vision exam to NDT personnel?

- a. It must be done by a procedure and performed by personnel approved by a Level III
- b. It must be done by a procedure approved by a Level III and must be administered by a licensed optometrist
- c. Qualifications and procedures for the vision exam are not addressed in CP-189
- d. It must be done by personnel approved by a Level III, but the procedure must be approved by a medical professional

6.1.4.

69. According to ANSI/ASNT CP-189, employer recertification for Level I/II is done:

- a. By completing General, specific and practical examinations on a 5 year interval
- b. By having 350 hours of documented experience and passing the specific for the first recertification interval, followed by accomplishing General, Specific and Practical examinations at the subsequent recertification interval (10 years).
- c. Continued satisfactory performance
- d. Completion of only those exams deemed necessary by the Level III
- e. A and B are correct

8.1 and 6.3; 8.1.1 and 8.1.2

70. According to ANSI/ASNT CP-189, an inspector's certification expires:

- a. upon termination of employment
- b. at the end of 5 years for Level I/II
- c. if the inspector's ASNT NDT Level III certificate has expired
- d. All of the above

7.1

71. According to ANSI/ASNT CP-189, the education, skills, training, knowledge, and experience required for personnel to properly perform NDT defines:

- a. Training
- b. Experience
- c. Qualification
- d. Certification

2.1.22

72. What methods are in ANSI/ASNT CP-189, but not SNT-TC-1A?

- a. Electromagnetics, Guided Wave and Laser
- b. Neutron Radiography and Vibrations
- c. Laser and Vibrations
- d. Vibrations and Magnetic Flux Leakage
- e. All methods in CP-189 are listed in SNT-TC-1A

SNT-TC-1A Table 6.3.1 A and ANSI/ASNT CP-189 Appendix A

73. According to SNT-TC-1A, who is responsible for reviewing and approving a written practice?

- a. Employer's Level III
- b. Employer
- c. Customer
- d. Customer's Level III

SNT-TC-1A 5.5

BASIC LEVEL III PRACTICE EXAM

74. Which of the following methods are in SNT-TC-1A, but not CP-189?

- a. Electromagnetics, Ground Penetrating Radar, and Magnetic Flux Leakage
- b. Ground Penetrating Radar, Guided Wave, Microwave, and Laser
- c. Laser, Magnetic Flux Leakage, and Guided Wave
- d. Leak, Acoustic Emission and Radiological

SNT-TC-1A Table 6.3.1 A and CP-189 Appendix A

75. According to SNT-TC-1A, a film interpreter practical examination should include _____.

- a. Review and grading of a sufficient number of images
- b. A mockup of an X-ray setup and review of at least 2 images
- c. A mockup of an X-ray or Gamma ray setup and review of at least 2 images
- d. The practical is not required for Level II Limited certifications

8.5.2.2

76. According to SNT-TC-1A, exceptions to the recommendations in the document _____.

- a. Must be clearly delineated within the body of the document along with technical rational or justification for the deviation
- b. Do not have to be identified in document
- c. Should be identified in an Annex to the written practice with technical justification or rational supporting the deviation
- d. Exceptions are not permitted under any circumstances

1.4 last sentence

77. According to SNT-TC-1A, a practical exam is _____.

- a. Recommended for reevaluation of Level I and II personnel
- b. Mandatory for recertification of Level I and II personnel
- c. Mandatory for recertification of Level I, II, and III personnel
- d. Not required for initial certification

10.2, 8.5 and 12.1

78. According to SNT-TC-1A, what is the minimum level of certification that can perform NDT without written instruction or supervision?

- a. Trainee
- b. Level I
- c. Level II
- d. Level III

4.3.2

79. A candidate scored 26 of 35 questions on the General and 37 of 40 questions on Specific exam. According to SNT-TC-1A which of the following would be the minimum needed to pass the practical exam?

- a. Get 21 out of 30 items correct
- b. Get 22 out of 30 items correct
- c. Get 45 out of 50 items correct
- d. Get 24 out of 30 items correct

8.1.5

80. Use of outlines in CP-105 is _____ according to SNT-TC-1A.

- a. Mandatory
- b. Recommended
- c. Mandatory unless technical rational is stated in written practice for not using them
- d. Mandatory, but additional topics may be added at the discretion of the NDT Level III

7.3

BASIC LEVEL III PRACTICE EXAM

81. For Level I and II, the topics in CP-105 are _____ according to ANSI/ASNT CP-189.

- a. Mandatory and no other topics shall be included
- b. Recommended
- c. Mandatory unless technical rational is stated in written practice for not using them
- d. Mandatory, but additional topics may be added at the discretion of the NDT Level III

4.1.1

82. For Level III training, the topics in CP-105 are _____ according to ANSI/ASNT CP-189.

- a. Listed for information as no additional training is required beyond Level II
- b. Recommended
- c. Mandatory unless technical rational is stated in written practice for not using them
- d. Mandatory, but additional topics may be added at the discretion of the NDT Level III

4.1.1 and 4.1.4

83. Which of the following is true regarding CP-105?

- a. The outlines for Level I and II are not progressive, meaning successful completion of Level II training negates the need for Level I training
- b. Time spent on each topic shall be determined by the NDT Level III
- c. For Level I and II topics may be deleted when used with CP-189
- d. For Level I and II topics may not be expanded when used with CP-189
- e. The sequence that topics are covered shall strictly follow the outline

4.1.1

84. Legal aspects or liability for certification is retained by:

- a. The certificate holder
- b. The employee
- c. ASNT
- d. The employer

SNT-TC-1A, 9.1 and 9.3

85. Under what circumstances are the recommendations of SNT-TC1-A mandatory?

- a. For all military contracts
- b. For refinery work performed in foreign countries
- c. Unless otherwise specified in NDT contracts
- d. When agreed to in a contract

SG Question 1-13

86. A person with a four year B.A. degree in Drama will require _____ years of experience as a Level II to qualify for Level III in the method.

- a. 1
- b. 2
- c. 4
- d. 10

SNT-TC-1A 6.3.2

87. How many hours of training are required for Halogen Diode Leak Testing Level I?

- a. 2
- b. 12
- c. 24
- d. 40

CP-189 Appendix A and Table 6.3.1 a in SNT-TC-1a

PART B – MATERIALS, FABRICATION AND PRODUCT
TECHNOLOGY

1. Alloy steel can be described as:

- a. ordinary carbon steel to which has been added elements such as chromium, nickel, etc.
- b. steel which does not contain significant amounts of alloying elements other than carbon
- c. steels whose constituents remain in solution with each other at room temperature, non-magnetic and corrosion resistant
- d. an alloy containing tin, lead or antimony, and about 77 percent Cu obtained from the smelting of sulfide copper ores

N88, para 2; N89, para 3

2. Austenitic steels can be described as:

- a. ordinary carbon steel to which has been added elements such as chromium, nickel, etc.
- b. steel which does not contain significant amounts of alloying element other than carbon
- c. steels whose constituents remain in solution with each other at room temperature, non-magnetic and corrosion resistant
- d. an alloy containing tin, lead or antimony, and about 77 percent Cu obtained from the smelting of sulfide copper ores

N90, para 4; old 53, rt col

3. Carbon steels can be described as:

- a. ordinary carbon steel to which has been added elements such as chromium, nickel, etc.
- b. steel which does not contain significant amount of alloying elements other than carbon
- c. steels whose constituents remain in solution with each other at room temperature, non-magnetic and corrosion resistant
- d. an alloy containing tin, lead or antimony, and about 77 percent Cu obtained from the smelting of sulfide copper ores

N86, last to 87 first

BASIC LEVEL III PRACTICE EXAM

4. Changes in properties of a metal or alloy that generally occur slowly at room temperatures and more rapidly at elevated temperatures, is known as:

- a. Macroscopic
- b. Alloy
- c. aging
- d. anisotropy
- e. activation

N50; N90 bottom

5. A substance having metallic properties and being composed of two or more chemical elements of which at least one is an elemental metal, is known as:

- a. an alloy
- b. a chemical
- c. a steel
- d. a composite

N79, para 1

6. The term used to denote that a structure is visible at magnifications from one to ten diameters is:

- a. Microscopic
- b. Macroscopic
- c. Submicroscopic
- d. Molecular

Definitions of Metallurgical Terms page 31

7. In foundry practice, metal supports for holding cores in place within sand molds are known as:

- a. closed molds
- b. chills
- c. blind risers
- d. chaplets

N151, para 6

BASIC LEVEL III PRACTICE EXAM

8. Metal inserts imbedded in the surface of a sand core or mold or placed in a mold cavity to increase the cooling rate at that point are known as:

- a. Chills
- b. closed molds
- c. closed dies
- d. blind risers

N142, para 4

9. Forming or forging dies designed to restrict the flow of metal laterally to the cavity within the die set are known as:

- a. Open dies
- b. Chaplets
- c. Closed dies
- d. Hammer forge

Definitions of Metallurgical Terms, page 10; N179, 2nd para; N171, para 3-6; N172, Fig 7.1

10. Deviation from the main composition at a particular location in a solidified metal is referred to as:

- a. pipe
- b. inclusions
- c. segregation
- d. brittleness

N163, para 2

11. The loss of carbon from the surface of a ferrous alloy as the result of heating in a medium that reacts with the carbon at the surface is:

- a. Carburizing
- b. Burning
- c. Decarburization
- d. Deep etching

BASIC LEVEL III PRACTICE EXAM

12. Conditioning a metallic surface for examination at magnification of 10 diameters or less to reveal gross features such as segregation, cracks, porosity or grain flow, is done by:

- a. Grinding
- b. Deep etching
- c. Carburizing
- d. Decarburization

~N57, 1st

13. The forming of recessed parts by forcing the plastic flow of metal in dies and the reduction of tubing by pulling through a die, is known as:

- a. Piercing
- b. Drawing
- c. Extrusion
- d. Forging

N177; Old 144

14. The maximum stress to which a material may be subjected without any permanent strain remaining upon complete release of stress is:

- a. Ultimate yield
- b. Elastic limit
- c. Ultimate tensile
- d. Fatigue life

N71

15. The conversion of billet into lengths of uniform cross section by forcing the plastic metal through a die orifice of the desired cross-sectional outline is:

- a. Forging
- b. Drawing
- c. An extrusion
- d. A casting

N180

BASIC LEVEL III PRACTICE EXAM

16. Short, randomly distributed fissures in forged ferrous metals which are attributed to stresses produced by localized transformation and decreased solubility of hydrogen during cooling after hot working, are known as:

- a. Shrink
- b. Bursts
- c. Hot shortness
- d. Hot tears
- e. Flakes

CT-6-4, 7-31

17. A metal rupture formed at the surface during solidification of a casting because of uneven cooling at an area of transition between sections of varying thickness is called:

- a. Bursts
- b. Shrink
- c. Cold cracks
- d. Hot tears
- e. Fatigue cracks

N164; CT-6-4, 7-63

18. Metal brittleness in the hot forming range is:

- a. Bursts
- b. Hot tear
- c. Cold crack
- d. Hot shortness
- e. Fatigue cracks

N58; N288, para 4 Indirectly; Old 59, 1c, top

19. The base metal melted during welding as determined on the cross-section of the weld is the:

- a. Transformations zone
- b. Fusion zone
- c. Homogenization
- d. precipitation

N231,, Fig 8-36; N236, Fig 8-48

20. The base metal which has not been melted, but whose mechanical properties and microstructure has been altered by the heat of welding is known as the:

- a. Heat affected zone
- b. Fusion zone
- c. Modulation zone
- d. Transformation zone

N233, Fig 8-43; N237

21. An alloy may become hardened with the _____ of a constituent from a supersaturated solid solution.

- a. Solidification
- b. Post heating
- c. Precipitation
- d. Substitution

N50, 4th para

22. Excessive stress concentrations do not usually occur:

- a. at notches
- b. variable residual stresses are present
- c. if loading applied is different than design loading
- d. when changes in cross-section are gradual

N203, 8th para; N297, 1st

23. Which of the following is the most important item to be considered in the prevention of fatigue cracks?

- a. Favor rotating to minimize stress concentration
- b. Designed to minimize stress concentration
- c. Avoid shot peening
- d. Polish all metal surfaces

N297, 1st; N275, para 2

BASIC LEVEL III PRACTICE EXAM

24. Heating to and holding a suitable temperature and then cooling at a suitable rate for such purposes as reducing hardness, improving machinability, or obtaining desired mechanical, physical, or other properties is termed:

- a. Pre-heating
- b. Annealing
- c. Stress relieving
- d. Age hardening

N49, para 2-4; N58, para 2.8.4..2; Old 138, left col; 39 left col, para 2

25. The swerving or disturbance of an electric arc from its normal path because the residual magnetic forces is known as:

- a. Spatter
- b. Crater
- c. Arc short
- d. Lack of penetration
- e. Arc blow

N233 top; N232, 3rd; N231; ASM Definitions of Metallurgical Terms, Pg 2

26. A large forging has been repaired by welding. Magnetic particle inspection of the repair weld deposit reveals an irregular indication running parallel with the welds passes. It is located near the center of the bead and starts and ends by changing slightly from the axial direction. This kind of discontinuity is most likely a:

- a. Lap
- b. Lack of fusion
- c. Crack
- d. Crater
- e. Slag inclusion

New: Fig 8.33 and 34, 230, 231; 232, para 2 and 3, Fig 8-40

27. Porosity, misrun, cold shut, hot tears, and inclusions are examples of discontinuities found in:

- a. Forgings
- b. Ingots
- c. Castings
- d. Heat-treated parts
- e. Slag inclusion

N164; Old 81-86

28. The temperature of magnetic transformation below which a metal or alloy is magnetic and above which it is nonmagnetic, is called the:

- a. Melting temperature
- b. Curie point
- c. Ambient temperature
- d. Unstable temperature

N314, 1st para

29. A zone or area of minute discontinuities in castings occurring during solidification are called:

- a. Bursts
- b. Hot tears
- c. Flakes
- d. Fatigue cracks
- e. Shrinkage

N164

30. Which of the following best describes the cause of metal fatigue?:

- a. Loading beyond the yield strength of the material
- b. Impact loading
- c. Cyclic tensile loading
- d. Alternating compression loading
- e. Constantly increasing static loading

N76, last

31. Which of the following discontinuities is not associated with welding?

- a. Porosity
- b. Burst
- c. Lack of fusion
- d. Crater cracks
- e. Lack of penetration

N22, para 8.3.1; N230 to 232

32. Which of the following discontinuities is not associated with castings? :

- a. Hot tears
- b. Laminations
- c. Cold shut
- d. Microshrinkage
- e. Porosity

N164; N239 Fig 8-51

33. Severe grinding cracks appear as:

- a. Wavelike patterns
- b. Widely spaced cracks
- c. Lattice network patterns
- d. Star like patterns

34. Which of the following discontinuities usually appears in the form of a star with small cracks emanating from a central point? :

- a. First
- b. Crater crack
- c. Grinding cracks
- d. Heat-treated cracks
- e. Machine tears

N232, para 3; Old 119, left col, last para

35. Which of the following items listed below is a discontinuity found in ingots?:

- a. Laminations
- b. Bursts
- c. Nonmetallic inclusions
- d. Unfused chaplets

N175

36. Nonmetallic inclusions are found in:

- a. Sheet metal
- b. Bar stock
- c. Forgings
- d. Castings
- e. All the above

N371, N164, N230, 2nd para

37. Lamination is a discontinuity which is found in:

- a. Castings
- b. Welds
- c. Extrusions
- d. Sheet or plate material

N239, Fig 8-51 and N373, para 4

38. Crater cracks in welds may occur:

- a. At the end of the weld
- b. At the beginning of the weld
- c. At any location where welding is interrupted
- d. At all of the above

N232, 3rd; Old 119, left col, last

BASIC LEVEL III PRACTICE EXAM

39. Inherent discontinuities can be defined as:

- a. Discontinuities that are related to the cooling of the molten metal
- b. Discontinuities related to manufacturing processes such as grinding, forming, machine, etc.
- c. Baseline discontinuities
- d. Discontinuities formed during the initial making of the metal

CT-6-4, page 7-5; Check N7, 81, 85 and 163

40. Service discontinuities can be defined as:

- a. Discontinuities related to the manufacturing processes
- b. Acceptable flaws that propagate to an acceptable level
- c. Discontinuities related to the various service conditions such as stress, corrosion, fatigue and wear
- d. Depending on when discovered, any of the above

N376, last to N377

41. Discontinuities most likely to be found in hot rolled bar stock:

- a. Writing cracks
- b. Laminations
- c. Seams
- d. Fatigue cracks
- e. Bursts

ASM Definitions of Metallurgical Terms, page 50, LC

42. The most common failure mechanism which is associated with the sharp fillets, notches, or undercut is known as:

- a. Fatigue cracks
- b. Crystallization
- c. Shrink
- d. Any of the above depending on location

N76, para 2; Old 23, left, lower

43. Rapidly breaking the arc during welding is likely to cause: _____.

- a. Hot tear
- b. Cold crack
- c. Crater crack
- d. Slag inclusion

N232; Old Page 119, left, 4th para

44. Which of the following discontinuities are associated with forging?

- a. Lap and Burst
- b. Hot tear and Lap
- c. Misrun and Burst
- d. Hot tear and misrun

CT-6-4, page 7-8; N371 Implies answer

45. Which of the following welding processes would produce the smallest fusion zone?

- a. Gas Tungsten Arc
- b. Gas Metal Arc
- c. Shielded Metal Arc
- d. Electron Beam

N220, para 1 and 3; Old Page 111

46. Which process does not involve melting?

- a. Casting
- b. Powder Metallurgy
- c. Gas Metal Arc Welding
- d. Brazing

N189, para 3; Old Page 135

BASIC LEVEL III PRACTICE EXAM

47. Which of the following is not a solid state welding process?

- a. Diffusion bonding
- b. Forge Welding
- c. Diffusion welding
- d. Fusion welding

N198, para 5; N 203, 2nd and 4th; N221, para 3; Old Page 106-113

48. Devices that can be bonded onto a structure for determination of local deformation due to loading are called:

- a. Load frames
- b. Strain gages
- c. Optical comparators
- d. Striations

49. Stresses that remain in a part after cold working, welding, or heat treating are referred to as:

- a. Thermal stresses
- b. Residual stresses
- c. Residual fields
- d. Thermal fatigue stress

N: 50, para 3; 233, para 2; 234, para 4

50. Which of the following flaws are associated with weld joints?

- a. Hot tear, cold cracks, and porosity
- b. Porosity, underbead cracking, and slag inclusions
- c. Slag inclusions, bursts, and tungsten inclusions
- d. Misruns, overlap, undercut, underfill, and mismatch

N230; Old Pages 116-120

51. The process of using wax patterns for foundry molds is often referred to as:

- a. Permanenet mold casting
- b. Die casting
- c. Investment casting
- d. Centrifugal casting

N154, 1st, Old Page 92

52. Which of the following welding processes would not use a filler metal?

- a. EBW
- b. GMAW
- c. SMAW
- d. FCAW
- e. Electroslag

N220, para 2-3; N214 Fig 8.2.5 lists 3 that do; N224, para 3 Fig 8.29;
Old Page 111

53. Which of the following is an intentional welding discontinuity?

- a. Porosity
- b. Slugging
- c. Slag inclusions
- d. Crater crack

54. Forging laps could be caused by _____.

- a. Mismatch between cope and drag half of mold
- b. Forging at too low a temperature
- c. Inclusions
- d. Die mismatch

BASIC LEVEL III PRACTICE EXAM

55. EDM is stands for _____.

- a. Electrolytic Discharge Machining
- b. Electron Discharge Machining
- c. Educational Department of Metallurgy
- d. Electrical Discharge Machining

N259, para 2nd; Old 164

56. EDM is often used for _____.

- a. Welding without filler metal in a vacuum
- b. Making artificial flaws in NDT standards
- c. Melting molds
- d. Resistance welding sheet metal

N261, top

57. A porous oxide layer grown on the surface of aluminum alloys to prevent corrosion and increase wear resistance is referred to as _____.

- a. Plating
- b. Ion Vapor Deposition
- c. Thermal spray
- d. Anodizing

Definitions of Metallurgical Terms page 2; Old Page 180, right, last;
N280, para 4

58. The study of a material's resistance to cracking or crack growth is referred to as _____.

- a. Failure Analysis
- b. Fracture Mechanics
- c. POD
- d. Retirement for Cause

Old Page 7

59. The grain size distribution in a casting is typically _____.

- a. Equiaxed
- b. Preferentially oriented in the rolling direction
- c. Small at mold surface and columnar toward center
- d. Randomly oriented and homogenous throughout its volume

N160-61 (Fig 6-21); Old Page 82

60. A surface roughness measurement is made and the results are that the surface exceeds the minimum roughness requirements. A second measurement is made oriented 90 degrees to the first measurement and the reading is acceptable. What is the cause for the difference in the two readings?

- a. The measurement was faulty
- b. The measurement was across the lay instead of with the lay
- c. Operator error
- d. The surface was made smoother by the prior test of the surface

N247, para 5; Old Page 196, right column, para 4

61. A measuring device that measures accurately to the nearest 0.0001 inch is:

- a. Vernier caliper
- b. Vernier micrometer
- c. Ruler
- d. Optical comparator
- e. Tape measure

Old Pages 191-193.

62. Most honeycomb structures are usually joined by:

- a. Adhesive bonding
- b. Diffusion bonding
- c. Welding
- d. Hot Isostatic Pressing

N203, para 5; Old Page 163, left column, 4th para

BASIC LEVEL III PRACTICE EXAM

63. A cutting tool that is used in the industry that withstands temperatures over 1100 degree C (2000F) is:

- a. Diamond
- b. Ceramic
- c. Cemented carbide
- d. Carbon tool steel

N246, 3rd; Old Page 150, rt col, para 5

64. What type of work is performed at a Blooming Mill?

- a. Hot rolling
- b. Cold work
- c. Forming of cold rolled seamless pipe
- d. Casting

N173-74; Old Page 128-129

65. Which one applies to the casting process?

- a. Dendrites are formed first
- b. Grains are formed toward the hot zone
- c. The material is less subject to segregation
- d. The process does not involve melting

N160, para 2-4; Old Page 82, left col

66. The process of covering one metal with another metal for prevention of corrosion is:

- a. Cladding
- b. Annealing
- c. Anodizing
- d. Tempering

N96, para 1-2; Old Page 44, left col, 2nd para; page 58, rt col, 3rd para

67. What type of furnace produces the highest quality of steel?

- a. Bessemer converter
- b. Oxygen furnace
- c. Electric arc furnace
- d. Open hearth furnace

N83, 3rd, Old Page 49, rt coll, 1st para

68. What cast molding process is able to produce castings with sections as thin as 0.015 inch (1/64 inch)?

- a. Investment casting
- b. Permanent mold casting
- c. Die casting
- d. Sand casting

N157, para 2; Old Page 91, rt col, last para

69. Untreated water is stored in a 300 series stainless steel (CRES) pipe. The pipe is drained and a fluorescent penetrant inspection is performed on the inside diameter. The inspection reveals numerous indications most probably caused by:

- a. Stress corrosion cracking
- b. Hydrogen embrittlement
- c. Thermal fatigue
- d. Microbiological induced corrosion (MIC)
- e. Heat Affected Zone (HAZ) cracking

Not in book, N52

70. What would be a quick test to separate carbon steel from 300 series stainless (corrosion resistant) steel?

- a. Salt fog test
- b. Use a magnet
- c. Perform an electrical conductivity test
- d. Tensile test

N90, para 6 and 7

BASIC LEVEL III PRACTICE EXAM

71. What does cold rolling do to the ductility of a material?

- a. No effect
- b. Decreases
- c. Increases
- d. Restores it

N175 last to 176, 1st; Old Page 125, right column

72. An irregular white or light spot in the radiograph of a weld would most likely occur due to which welding process?

- a. ERW
- b. SMAW
- c. GMAW
- d. PAW

N230 last (fast?)

73. Which of the following would be considered a CFC?

- a. Difluoromethane
- b. 1-1-1 Carbon
- c. R-12
- d. 1-1-1 Trichlorotrifluoroethane
- e. c and d

N: 272, 1st and 3rd; 274 describes process application

74. Which of the following would be associated with hydrogen contamination in hot worked steel?

- a. Lamination
- b. Flake
- c. Cold shut
- d. Lap
- e. Burst

75. What metal is used for hot dip galvanizing?

- a. Cadmium
- b. Aluminum
- c. Zinc
- d. Gold

N281, para 5 and 6; Old 64, RC, 3rd ; 179, RC, 6th

76. Which of the following alloying elements would have a negative effect on the corrosion resistance of steel:

- a. Silicon
- b. nickel
- c. molybdenum
- d. sulfur

77. A dimension of 1.248 +/- 0.001 would be considered a

- a. bilateral tolerance
- b. unilateral tolerance
- c. limiting dimension
- d. convolution

78. What joining process uses temperatures above 800 F and does not require melting of the base metal?

- a. Adhesive Bonding
- b. Brazing
- c. Soldering
- d. Welding

N201, para 6; Old 98, Left

BASIC LEVEL III PRACTICE EXAM

79. Two main groups of plastics are:

- a. Hard and soft
- b. Thermosetting and thermoplastic
- c. Polymer and nonpolymer
- d. Fibrous and filled

N108, para 4 and 5; Old 67, right

80. The welding process designated GTAW utilizes an electrode known as tungsten. The electrode is _____.

- a. Consumable
- b. Nonconsumable
- c. A shielding gas emitter
- d. Flux cored consumable

N218, para 3

81. 18% chromium in stainless steel is most used because

- a. It's too hard
- b. Has high carbon
- c. it's inexpensive
- d. withstand high temperature

N89, last; N90, 3rd and 6th; Old 53, rt, 5

82. Of the following materials, which cannot be joined to carbon steel by brazing or welding?

- a. Cast iron
- b. Aluminum
- c. Nickel alloy
- d. Copper

83. In powder metallurgy parts are made by use of pressure, metal powder and _____.

- a. Sintering
- b. Ultrasonics
- c. Oxygen
- d. Melting

N189, para 3; N190, para 2; N191 para 2 (Best); Old 137, right

84. While performing penetrant inspection of an aluminum part, a linear indication is along the toe of a fillet weld. The indication is likely _____.

- a. Incomplete penetration
- b. Porosity
- c. Overlap
- d. Hot tear

N229, Fig 8.34; Old 117, Fig 10-20

85. The difference between die casting and permanent mold casting is _____.

- a. Die casting uses a sand mold
- b. There is no difference as the two terms are used synonymously
- c. Die casting uses ceramic molds
- d. Pressure is used to flow metal into the mold during die casting

N155, last para; Old 91, left, last para

86. Grain structure in a casting is strongest _____.

- a. Across the grain
- b. Parallel to the grain
- c. Perpendicular to the grain
- d. At a 45 degree angle to the grain

~N159, para 6

87. How many orders of magnitude is it going from micrometers to kilometers?

- a. 3
- b. 6
- c. 9
- d. 12

88. Crater cracks can occur when?

- a. When the arc is started
- b. At the end of the arc
- c. During welding in the midst of applying and removing the arc
- d. At any time the arc is removed too quickly

N232, 3rd; ASNT VT Hdbk 3rd ed, page 194, left col, para 2; Old M&P 119, left; ASNT NDT Hdbk Vol 10, 3rd ed, page 81, rt col

89. What causes undercut on a Weld?

- a. Excessive amperage
- b. Low voltage
- c. Low amperage
- d. Low travel speed
- e. Short Arc length
- f. Small electrode diameter

ASNT 3rd ed VT Hdbk, page 195, right col, 2nd para; description Old M&P 118, rt col; ASNT NDT Hdbk Vol 10, 3rd ed, page 82
N231, 2nd described as poor technique Fig 8.39